

EXAM REVIEW FOR MCV 4U1
VECTORS AND CALCULUS

Study Tips:

- *The best way to study for a math exam is to redo all Unit Tests.*
- *Write out all of your formulas and key concepts in one place to help you remember everything since no formulas will be given on the exam.*
- *Complete the following extra exercises*

Reminders:

- **DO NOT COME TO YOUR EXAM WITHOUT BRINGING YOUR TEXTBOOK TO RETURN**
- **Bring your own calculator, pencil, eraser and ruler to the exam (No sharing allowed)**
- **There are no formula sheets, questions or electronic dictionaries allowed during the exam.**

PART A: CALCULUS

Limits, Continuity, and First Principles:

1. Evaluate each limit, if it exists.

$$\begin{array}{lll} \text{a) } \lim_{x \rightarrow 3} \frac{x^2 - 9}{x^2 + 7x - 30} & \text{b) } \lim_{x \rightarrow 2} \frac{3\sqrt{x-1} - 3}{x - 2} & \text{c) } \lim_{x \rightarrow 2} \frac{\frac{1}{2} + \frac{1}{x-4}}{2x - 4} \\ \text{d) } \lim_{x \rightarrow 0} \frac{(x+64)^{\frac{1}{3}} - 4}{x} & \text{e) } \lim_{x \rightarrow \infty} \frac{x^2 - x^3 + 2x}{5x(x - 2x^2)} & \end{array}$$

2. I) Determine all discontinuities and state which conditions of continuity fail for each discontinuity.

$$\text{a) } f(x) = \frac{3x-6}{x^2+8x-20} \quad \text{b) } f(x) = \begin{cases} 3x^2-1, & x \geq 4 \\ 5x+10, & 0 < x < 4 \\ 3x^2+10, & x < 0 \end{cases}$$

$$\text{II) Find all values of } a \text{ and } b \text{ such that the function } f(x) = \begin{cases} 2x & \text{if } x \in (-\infty, a] \\ x^2 & \text{if } x \in (a, b) \\ 2x-1 & \text{if } x \in [b, \infty) \end{cases} \text{ is continuous for all values of } x.$$

3. Find the derivative of each function using **FIRST PRINCIPLES**.

$$\text{a) } f(x) = 3x^2 + 2x - 1 \quad \text{b) } f(x) = \frac{3}{2-x} \quad \text{c) } y = 4 - 2\sqrt{x+3}$$

Derivatives and Basic Applications

4. Find $\frac{dy}{dx}$ for each curve. (Do not simplify your final answer.)

$$\begin{array}{llll} \text{a) } y = 5x^{-4} + \frac{2}{x^{-\frac{3}{2}}} - 5\pi + x & \text{b) } y = \frac{\sqrt{3x^2-2}}{5x} & \text{c) } y = \sqrt[4]{e^{4x} + 4x} & \text{d) } y = x^7 e^{x^7} + 8^{x-1} \cdot 4^{-x^2 - \sec x^3} \end{array}$$

$$\begin{array}{llll} \text{e) } 3y^2 - 8x^2 y + 2x = 5 & \text{f) } 7xy^3 - 2xy = 3y^4 & \text{g) } y = (9 \log_3 x)^5 & \text{h) } y = \frac{7x^2 \tan^3(\ln x^2)}{\sqrt{1 + \cos^2 x}} \end{array}$$

$$\text{i) } y = \sin(\log(5x^2 + \pi^3)) + \cot^2(\cos(\ln(6x-4)))$$

5. Find the equation of the tangent at the given value.

a) $f(x) = \frac{3x^2 - 2}{x^4 - 2}$, at $x = 1$

b) $f(x) = (6x^3 + 2)(5x - 1)^4$, at $x = 0$

c) $3x - y = 2 \ln x$, at $x = 1$

d) $4x - e^{xy} = 2y$ at the y-intercept

6. If $f(x) = x^2 + ax + b$ has a horizontal tangent line at $(-3, -12)$, find the values of **a** and **b**.

Word Applications (Position, Velocity, Acceleration, Related Rates, and Optimization)

7. Determine the initial velocity and acceleration for the position function $s(t) = \frac{3t}{1+t^3}$ where s is in km and t is in minutes.

8. Given the horizontal position function, $s(t) = 2t^2 e^{-t}$, where s is in meters and t is time in seconds, find:

a) When is the object at rest? What is its position in terms of the fixed point? (Be specific)

b) When is the object moving left?

c) When is the velocity constant?

d) time(s) at which acceleration is equal to zero?

e) Is the object speeding up or slowing down at 4 seconds?

9. A 15 meter ladder is resting against the wall. The bottom is initially 10 meter away from the wall and is being pushed towards the wall at a rate of $\frac{1}{4}$ m/sec. How fast is the top of the ladder moving up the wall 12 seconds after we start pushing? [Ans: $\frac{7}{4\sqrt{176}}$ m/sec]

10. A tank of water in the shape of a cone is leaking water at a constant rate of $2\text{m}^3/\text{hr}$. The base radius of the tank is 5 meter and the height of the tank is 14 meter.

(a) At what rate is the depth of the water in the tank changing when the depth of the water is 6 m?
[Ans: 0.1386 m/hr]

(b) At what rate is the radius of the top of the water in the tank changing when the depth of the water is 6 m?
[Ans: -0.04951 m/hr]

11. Two people on bikes are separated by 350 meters. Person A starts riding north at a rate of 5 m/sec and 7 minutes later Person B starts riding south at 3 m/sec. At what rate is the distance separating the two people changing 25 minutes after Person A starts riding? [Ans: 7.9958 m/sec.]

12. A cylindrical bread role is growing at $3\text{ cm}^3/\text{min}$ and is increasing in length at $1\text{ cm}/\text{min}$. Determine the rate of change of the radius when the length of the role is 10 cm and the radius is 4cm.

13. A spherical balloon is being filled such that the circumference is increasing at $4\pi\text{cm/s}$. Determine the rate at which the volume of the balloon is changing when the circumference is $24\pi\text{cm}$.
14. A water tank in the shape of an inverted cone is draining at $2\text{m}^3/\text{min}$. If the tank has a height of 10 meters and a diameter of 6 meters, find the rate at which the water level is falling when the water is 1.3 meters deep. (Round your answer to 2 decimal places)
15. What is the maximum volume of a cylinder with a surface area of 142 cm^2 ? (Give your answer to the nearest cm^3)
16. Given that a square based box has a volume of 2.8 m^3 . Find the dimensions of the box that will minimize the cost of the material to make the box knowing that the top and bottom of the box are FOUR times as expensive as the sides. (Round your answer to two decimal places.)
17. A piece of string 30 cm long is cut into 2 pieces to make a square and a circle. Determine the lengths of each piece of string in order to minimize the total area of both shapes. (Round to one decimal place.)
18. Find the largest volume of a cylinder that can be inscribed in a sphere with a radius of 9cm.
19. Find the absolute maximum and the minimum value on the interval $-3 \leq x \leq 3$ for each function.
- a) $f(x) = x^3 + 3x^2 + 3x - 7$ b) $f(x) = \frac{x-3}{x+5}$ c) $f(x) = \frac{6x}{1+x^2}$
20. Find the absolute maximum and the minimum value on the given interval:
- a) $f(x) = 2\cos(x^2)$ on $0 \leq x \leq 2\pi$ b) $y = \frac{1}{4}x^2(2\ln x - 3)$ on $1 \leq x \leq 3$ c) $g(x) = x^2e^x$ on $-3 \leq x \leq 1$

Curve Sketching:

21. Find all intervals of increase and decrease and local max and min values for each function:

a) $f(x) = x^3 - 9x^2 + 24x - 15$ b) $f(x) = \frac{x^2 - x - 1}{x - 1}$

22. Use the second derivative test on $f(x) = 2x^3 - 9x^2 + 9$ to determine if the critical values are maximum or minimum points.

23. Determine any vertical, horizontal or oblique asymptotes for each function:

a) $f(x) = \frac{x^3 - 4x^5 + 2x}{x^2(x^3 - 8)}$ b) $f(x) = \frac{5x^2 - 11x + 1}{\ln(x^2) - 3}$ c) $f(x) = \frac{5x}{e^{x^2} - 3}$

24. Determine intercepts (where possible), asymptotes, critical points, intervals of increase/decrease, local max/min points, intervals of concavity, and points of inflection, and then sketch each function. (Show all work)

a) $f(x) = \frac{1+x^2}{1-x^2}$, given that $f''(x) = \frac{12x^2 + 4}{(1-x^2)^3}$ b) $f(x) = \frac{x^2 - x + 2}{x - 2}$, given that $f''(x) = \frac{8}{(x-2)^3}$

25. Find the constants a, b and c so that the graph of $f(x) = 4x^3 + ax^2 + bx + c$ has a relative maximum at (1, 2) and point of inflection at $x = -1$.

26. Sketch a function with the following properties:

- a) $\lim_{x \rightarrow \pm\infty} f(x) = 3$
- b) $\lim_{x \rightarrow -2} f(x) = \pm\infty$ [no limit] and $\lim_{x \rightarrow 2} f(x) = \pm\infty$ [no limit]
- c) $f'(x) < 0$ for $x < 0$ only
- d) $f''(x) > 0$ for $-2 < x < 2$ only
- e) $f(-4) = -3$ and $f(4) = -2$

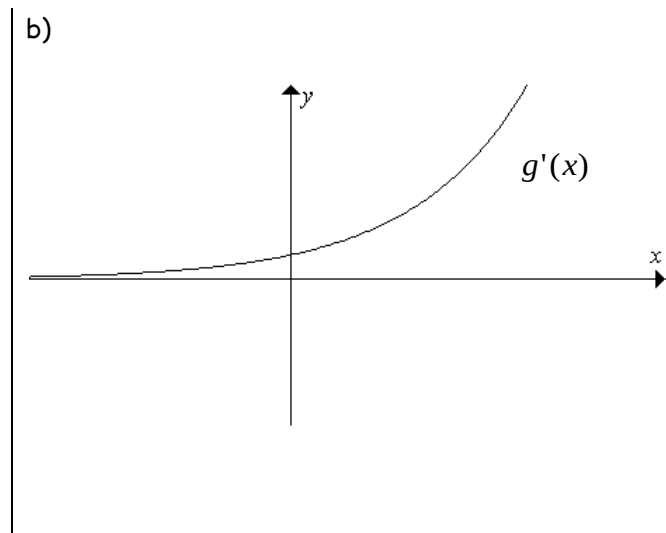
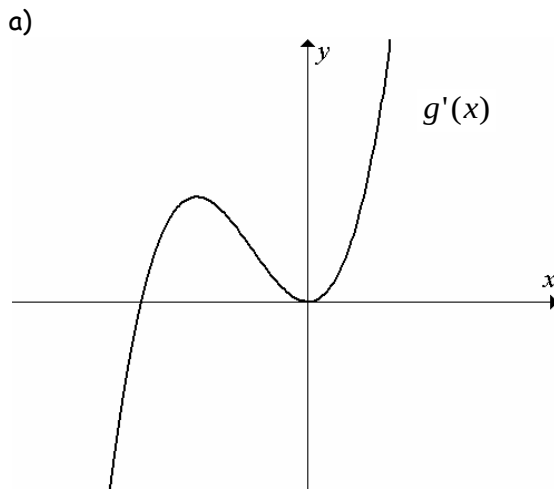
27. Sketch the graph of a polynomial function that satisfies the following conditions:

- ☞ $f(x) \leq 0$ when $x \leq -3$ and when $x \geq 0$
- ☞ first differences are positive when $x < -2$
- ☞ second differences are negative when $x < -1$ and when $x > 0$

28. Create a graph with all of the following characteristics. Determine an equation of a function that models your graph.

- | | |
|--|---|
| $f'(x) > 0$ for $x < -2$ and $-2 < x < 0$ | $f'(x) < 0$ for $0 < x < 3$ and $x > 3$ |
| $f''(x) > 0$ for $x < -2$ and $x > 3$ | $f''(x) < 0$ for $-2 < x < 3$ |
| $\lim_{x \rightarrow \pm\infty} f(x) = 2$ | $\lim_{x \rightarrow 3^-} f(x) = -\infty$ |
| $\lim_{x \rightarrow 3^+} f(x) = \infty$ | $\lim_{x \rightarrow -2^-} f(x) = \infty$ |
| $\lim_{x \rightarrow -2^+} f(x) = -\infty$ | $f(0) = 0$ |

29. Given the graph of the first derivative of the function $g(x)$, sketch the graph of the function and of the second derivative.



30. Analyse each of the following functions and use your analysis to sketch the graph of the function.

a) $f(x) = \frac{5x}{(x-1)^2},$	$f'(x) = \frac{-5(x+1)}{(x-1)^3},$	$f''(x) = \frac{10(x+2)}{(x-1)^4}$
b) $g(x) = (x-1)^3 - 1,$	$g'(x) = 3(x-1)^2,$	$g''(x) = 6(x-1)$
c) $k(x) = \sqrt{x+5},$	$k'(x) = \frac{1}{2\sqrt{x+5}},$	$k''(x) = \frac{-1}{4(x+5)^{\frac{3}{2}}}$
d) $j(x) = (x-1)^{\frac{1}{3}} + 2,$	$j'(x) = \frac{1}{3}(x-1)^{-\frac{2}{3}},$	$j''(x) = \frac{-2}{9}(x-1)^{-\frac{5}{3}}$
e) $h(x) = x\sqrt{9-x^2},$	$h'(x) = \frac{9-2x^2}{\sqrt{9-x^2}},$	$h''(x) = \frac{2x^3 - 27x}{(9-x^2)^{\frac{3}{2}}}$
f) $m(x) = \frac{(x-1)^2}{(x+1)^3},$	$m'(x) = \frac{-x^2 + 6x - 5}{(x+1)^4},$	$m''(x) = \frac{2(x^2 - 10x + 13)}{(x+1)^5}$

PART B: VECTORS

Geometric Representation

31. Determine the magnitude of the resultant vector of two forces of 54 N and 34 N acting at an angle of 55° to each other.
32. Two forces at an angle of 130° to each other act on an object. Determine their magnitudes if the resultant has a magnitude of 480 N and makes an angle of 55° with one of the forces.
33. The pilot of an airplane that flies at 800 km/h wishes to travel to a city 800 km due east. There is a 80 km/h wind from the north east.
- What should the plane's heading be?
 - How long will the trip take?
34. A 10 kg weight is supported by two strings of length 5 m and 7 m attached to two points in the ceiling 10 m apart. Find the tension in each string.

Algebraic Representations

35. Which of the following vectors are unit vectors? Explain.

$$\vec{a} = \left[\frac{2}{3}, \frac{1}{3}, \frac{2}{3} \right] \quad \vec{b} = \left[\frac{1}{\sqrt{26}}, \frac{4}{\sqrt{26}}, \frac{-3}{\sqrt{26}} \right] \quad \vec{c} = \left[\frac{3}{4}, \frac{-\sqrt{3}}{4}, \frac{1}{2} \right] \quad \vec{d} = \left[\frac{-11}{2}, 4, \frac{7}{2} \right]$$

36. Calculate the scalar and vector projection of $\vec{a}$ onto $\vec{b}$ given that:

$$\text{a) } \vec{a} = [2, 3] \text{ and } \vec{b} = [-1, 1] \quad \text{b) } \vec{a} = [0, 2, 1] \text{ and } \vec{b} = [4, 6, -1]$$

37. Points A(2,4), B(0,0) and C(-2,1) define a triangle in a plane. Find angle ABC to the nearest degree.

38. Consider two lines with equations $\frac{x+8}{1} = \frac{y+4}{3} = \frac{z-2}{1}$ and $[x, y, z] = [3, 3, 3] + t[4, -1, -1]$.

- Show that the two line are perpendicular.
- Find the point of intersection between the lines.

39. Find the vector, parametric, and symmetric equations of the line (if possible):

- that passes through the point (0, -1, 2) and in parallel to the line $x = 3 + 2t, y = 5t, z = -1 - t, t \in \mathbb{R}$.
- That passes through points (0, 0, 1) and (1, 0, 0).

40. Find the Cartesian equation of the plane that:

- passes through the points (3, 2, 3), (-4, 1, 2) and (-1, 3, 2)
- passes through the point (1, 2, 4) and is parallel to the plane $y + z = 5$
- contains the lines $\vec{r} = [2, 1, 7] + t[0, 1, 0]$ and $\vec{r} = [3, 0, 4] + t[2, -1, 0]$

41. For what values of k, if any, will the planes $3x + ky + z - 6 = 0$ and $6x + (1-k)y + 2z - 9 = 0$ be:

- Parallel
- Perpendicular

42. Find the shortest distance between the:

- point (1, 3, 2) and the line $\frac{x-1}{-1} = \frac{y-3}{1} = \frac{z-7}{2}$
- point (3, 2, 1) and the plane $3x + 2y + z = 10$
- the planes $x + 2y - 5z - 10 = 0$ and $2x + 4y - 10z - 17 = 0$

43. Determine the intersection (if any) between the following sets of planes. Give a geometrical interpretation for each situation. Explain how many solutions each situation has (one unique solution, infinite solutions or no solution).

- | | | | |
|--------------------------|------------------------|------------------------|-----------------------|
| a) $3x - y + 4z - 7 = 0$ | $2x + 4y + z = 2$ | $x + 2y + 3z = 4$ | $x - 3y + 2z = -3$ |
| $x + y - 2z + 5 = 0$ | b) $5x + 5y + 3z = 17$ | c) $3x + 6y + 9z = 12$ | d) $3x - 4y + z = 11$ |
| | $4x - y + 3z = 26$ | $3x + 5y - 2z = 1$ | $8x - 9y + z = 5$ |

Answers to Exam Review for MCV 4U1

Answers:

1. a) $\frac{6}{13}$ b) $\frac{3}{2}$ c) $\frac{-1}{8}$ d) $\frac{1}{48}$ e) $\frac{1}{10}$

2. I) a) Discontinuous at $x = 2$ with a hole, conditions 1 and 2 fail
Discontinuous at $x = 10$ with a vertical asymptote, conditions 1, 2 and 3 fail.
b) Discontinuous at $x = 4$ with a break, conditions 2 and 3 fail.
Continuous at $x = 0$.

II) $a = 0$; $b = 1$

3. a) $f'(x) = 6x + 2$ b) $f'(x) = \frac{3}{(2-x)^2}$ c) $\frac{dy}{dx} = \frac{-1}{\sqrt{x+3}}$

4. a) $\frac{dy}{dx} = -20x^{-5} + 3\sqrt{x} + 1$ b) $\frac{dy}{dx} = \frac{15x^2(3x^2-2)^{-\frac{1}{2}} - 5(3x^2-2)^{\frac{1}{2}}}{25x^2}$ c) $\frac{dy}{dx} = \frac{1}{4}(e^{4x} + 4x)(4e^{4x} + 4)$
d) $\frac{dy}{dx} = 7x^6 e^{x^7} + 7x^{13} e^{x^7} + 8x^{-1} \ln 8 \left(4^{-x^2} - \sec x^3 \right) + 8x^{-1} \left(4^{-x^2} - \sec x^3 \right) (-2x - 3x^2 \sec x^3 \tan x^3) \ln 4$
e) $\frac{dy}{dx} = \frac{8xy-1}{3y-4x^2}$ f) $\frac{dy}{dx} = \frac{2y-7y^3}{21xy^2-2x-12y^3}$ g) $\frac{dy}{dx} = \frac{45(9\log_3 x)^4}{x \ln 3}$

h) $y = 7x^2 \tan^3(\ln x^2) (1 + \cos^2 x)^{-\frac{1}{2}}$
 $\frac{dy}{dx} = 14x \tan^3(\ln x^2) (1 + \cos^2 x)^{-\frac{1}{2}} + (7x^2) 3 \tan^2(\ln x^2) \sec^2(\ln x^2) \left(\frac{2x}{x^2} \right) (1 + \cos^2 x)^{-\frac{1}{2}}$
 $- \frac{1}{2} (1 + \cos^2 x)^{-\frac{3}{2}} (-2 \sin x \cos x) 7x^2 \tan^3(\ln x^2)$

i) $\frac{dy}{dx} = \frac{10x}{(5x^2 + \pi^3) \ln 10} \cdot \cos(\log(5x^2 + \pi^3)) + 2 \cot(\cos(\ln(6x-4))) \frac{6}{6x-4} (-\sin(\ln(6x-4))) \cdot (-\csc^2(\cos(\ln(6x-4))))$

5. a) $y = -10x + 9$ b) $y = -40x + 2$ c) $y = x + 2$ d) $y = \frac{9}{4}x - \frac{1}{2}$

6. $a = 6$ and $b = -3$

7. $v(0) = 3 \text{ km/min}$

8. Velocity:

Acceleration :

$$v(t) = 4te^{-t} - 2t^2e^{-t}$$

$$a(t) = 2e^{-t}(t^2 - 4t + 2)$$

$$= 2te^{-t}(2-t)$$

$$a(0) = 4 \text{ km/min}$$

$$v(t) = 0 \Rightarrow t = 0, t = 2$$

a) at rest at origin and 2 second; position: $\frac{8}{e^2} \doteq 1.1 \text{ m}$ to the right of the fixed point

b) Moving left means $v(t) < 0$, moving left when $t \in (2, \infty)$ seconds

c& d) velocity is constant when $a(t) = 0$; it occurs when $t = 3.4s, t = 0.58s$

e) $v(4) < 0, a(4) > 0 \rightarrow v(4)a(4) < 0$, $\therefore$ it is slowing down at 4 seconds

9. Answer is given

10. Answer is given

11. Answer is given

12. $\frac{dr}{dt} = \frac{3-16\pi}{80\pi} \text{ cm/min}$

13. $\frac{dV}{dt} = 1152\pi \text{ cm}^3/\text{s}$ (Check)

14. $\frac{dh}{dt} = \frac{13.15}{\pi} \doteq -4.19 \text{ m/min}$

15. $V = 130\text{cm}^3$

16. $\text{length} = \text{width} \doteq 0.89\text{m}$, $\text{height} \doteq 3.55\text{m}$

17. Use 13.2 cm for the circle and 16.8 cm for the square.

18. $V = 324\pi\sqrt{3} \text{ cm}^3$

19. a) absolute max value is 56, absolute min value is -16

b) absolute max value is 0, absolute min value is -3

c) absolute max value is 3, absolute min value is -3

20. a) absolute max value is $2\cos(\sqrt{n\pi})^2 = 2$; $n = 0, 2, 4$, absolute min value is $2\cos(\sqrt{n\pi})^2 = -2$; $n = 1, 3$

b) absolute max value is $\frac{-3}{4}$, absolute min value is $\frac{-1}{4}e^2$ c) absolute max value is e , absolute min value is 0

21. a) Inc: $x \in (-\infty, 2) \cup (4, +\infty)$, Dec: $x \in (2, 4)$ b) Inc: $x \in (-\infty, 1) \cup (1, +\infty)$, Dec: Never

22. $f''(0) < 0 \rightarrow \text{concave down} \rightarrow \text{max point} = (0, 9)$, $f''(3) > 0 \rightarrow \text{concave up} \rightarrow \text{min point} = (3, 36)$

23. a) VA: $x = 2$, $x = 0$; HA: $y = -4$; OA: none

b) VA: $x = \pm\sqrt{e^3}$; HA: none; OA: none

c) VA: $x = \pm\sqrt{\ln 3}$, HA: $y = 0$, OA: none

24. a) $f(x) = \frac{1+x^2}{1-x^2}$, $f'(x) = \frac{4x}{(1-x^2)^2}$, $f''(x) = \frac{12x^2+4}{(1-x^2)^3}$

$$D = \{x \neq \pm 1, x \in \mathbb{R}\}$$

$$V.A: x = \pm 1 \rightarrow \lim_{x \rightarrow 1^-} f(x) = \infty$$

$$\lim_{x \rightarrow 1^+} f(x) = -\infty$$

$$\lim_{x \rightarrow -1^-} f(x) = -\infty$$

$$\lim_{x \rightarrow -1^+} f(x) = \infty$$

$$H.A: y = -1 \rightarrow \lim_{x \rightarrow \infty} f(x) = -1 (\text{below})$$

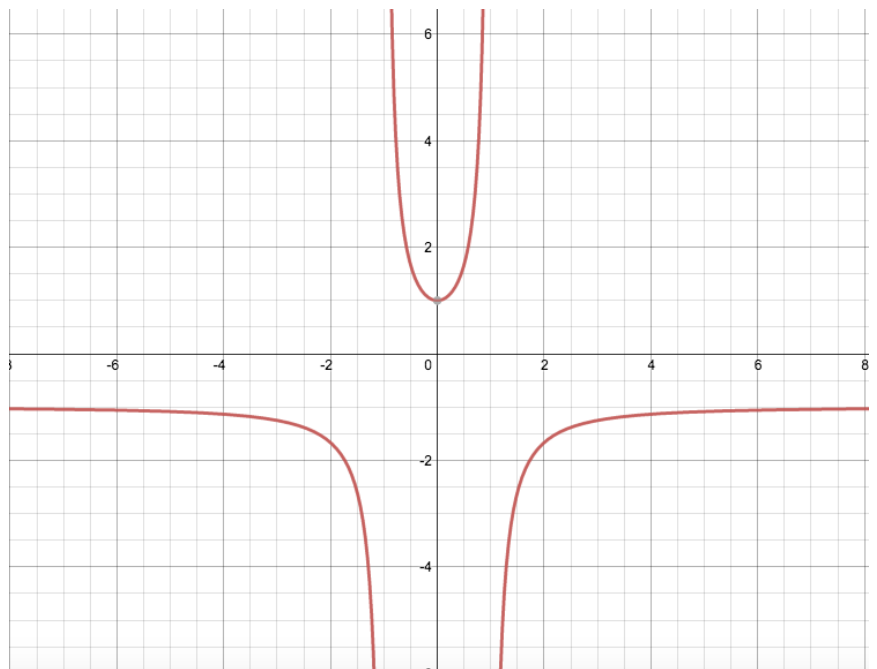
$$\lim_{x \rightarrow -\infty} f(x) = -1 (\text{below})$$

$$f'(x) = 0: x = 0$$

local min $(0, 0)$

$$f''(x) > 0 \rightarrow f \in (-1, 1)$$

$$f''(x) < 0 \rightarrow f \in (-\infty, -1) \cup (1, \infty)$$



25.

$$f(x) = 4x^3 + ax^2 + bx + c$$

$$f'(x) = 12x^2 + 2ax + b$$

$$f''(x) = 24x + 2a$$

$$f(1) = 2 \rightarrow 4 + a + b + c = 2$$

$$f'(1) = 0 \rightarrow 12 + 2a + b = 0$$

$$f''(-1) = 0 \rightarrow -24 + 2a = 0 \rightarrow \boxed{a = 12}$$

$$2a + b = -12$$

$$b = -12 - 24$$

$$\boxed{b = -36}$$

$$a + b + c = -2$$

$$12 - 36 + c = -2$$

$$\boxed{c = 22}$$

26.

$$1) \lim_{x \rightarrow \pm \infty} f(x) = 3 \Rightarrow \text{HA } y=3$$

$$\lim_{x \rightarrow -2} f(x) = \pm \infty \Rightarrow \text{VA } x=-2 \text{ (single root)}$$

(arrows in opposite directions)

$$\lim_{x \rightarrow 2} f(x) = \pm \infty \Rightarrow \text{VA } x=2$$

"

$$f'(x) < 0 \text{ for } x < 0 \text{ only} \Rightarrow f(x) \text{ decr. when } x < 0$$

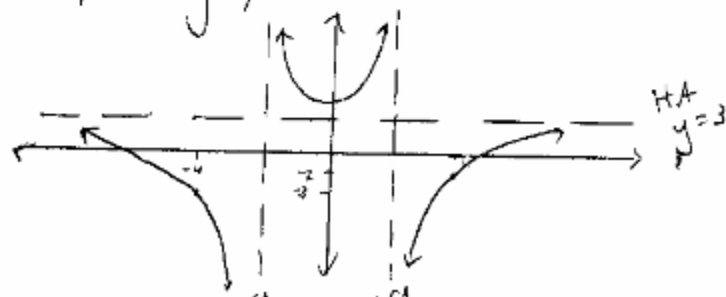
incr. elsewhere

$$f''(x) > 0 \text{ for } -2 < x < 2 \text{ only} \Rightarrow \text{conc. up when } -2 < x < 2$$

conc. down elsewhere

$$f(-4) = -3 \text{ and } f(4) = -2$$

pts on graph.



27.

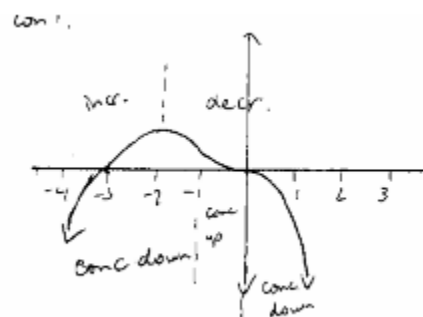
$$f(x) \leq 0 \text{ when } x \leq -3 \text{ and when } x \geq 0$$

y values are -ve or zeros!

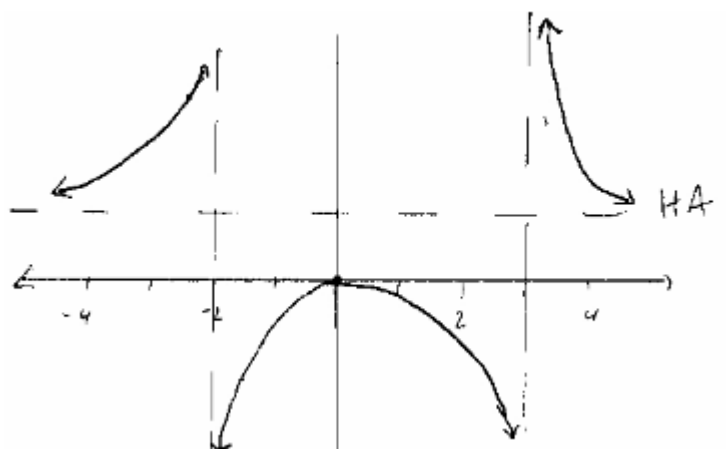
$$\Delta y > 0 \text{ when } x < -2 \Rightarrow f(x) \text{ is incr.}$$

$$\Delta^2 y < 0 \text{ when } x < -1 \text{ and when } x > 0$$

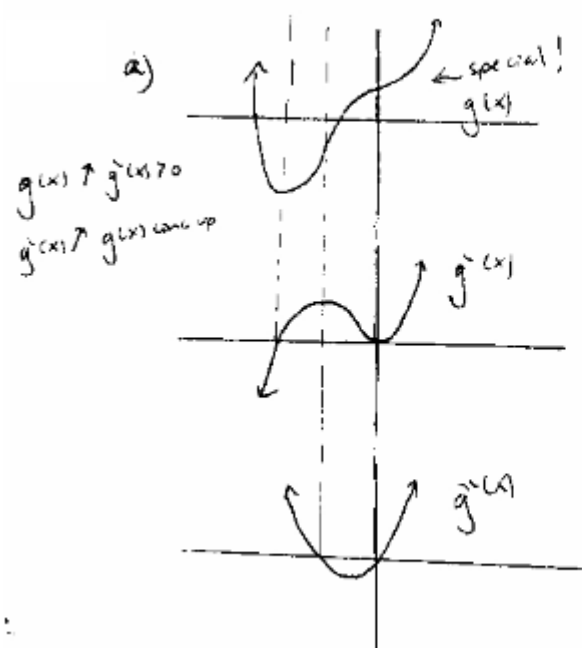
$\Rightarrow f(x)$ is concave down.



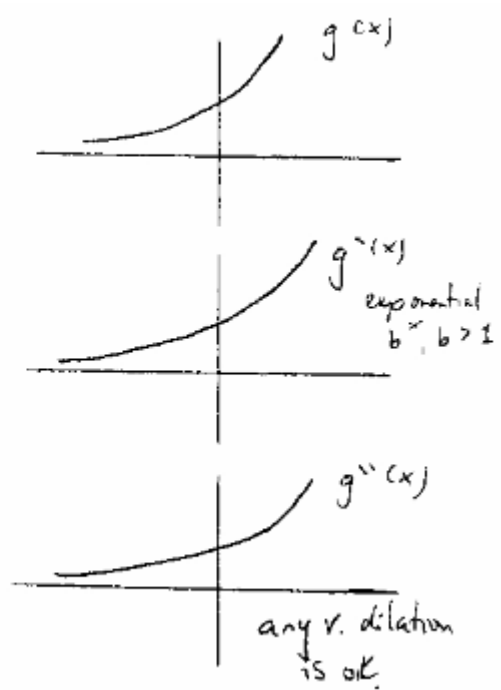
28.



29.



b)



30) Analyse each of the following functions and use your analysis to sketch the graph of the function.

a) $f(x) = \frac{5x}{(x-1)^2}$, $f'(x) = \frac{-5(x+1)}{(x-1)^3}$, $f''(x) = \frac{10(x+2)}{(x-1)^4}$

D: $\{x \mid x \neq -1, x \in \mathbb{R}\}$

Zero: 0,

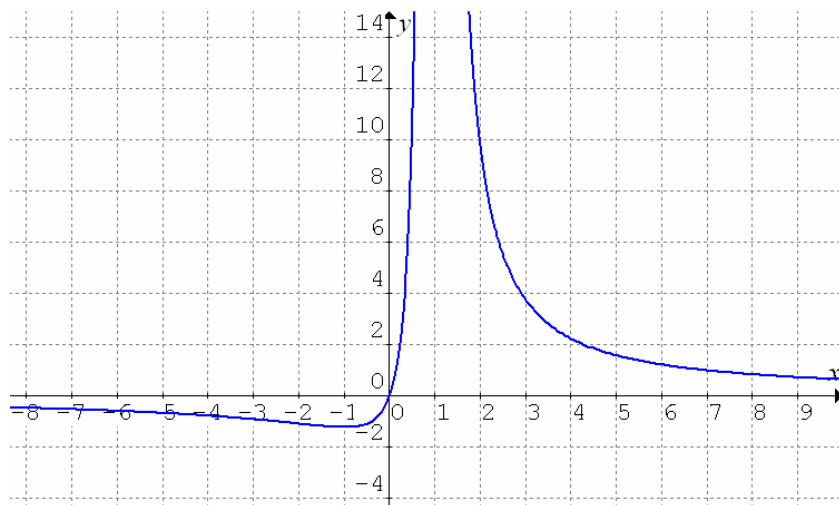
y-int: 0

VA: $x = 1$,

HA: $y = 0$

Min (-1, -1.25)

Pt. of infl. (-2, -1.11)



b) $g(x) = (x-1)^3 - 1$,

$g'(x) = 3(x-1)^2$,

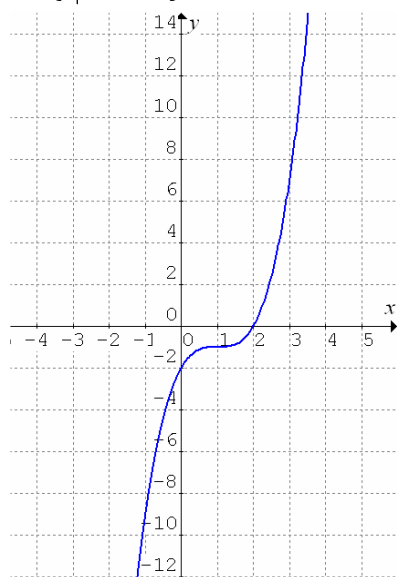
$g''(x) = 6(x-1)$

D: $\{x \mid x \in \mathbb{R}\}$

Zero: 2,

y-int: -2

Pt. of infl. (1, -1)

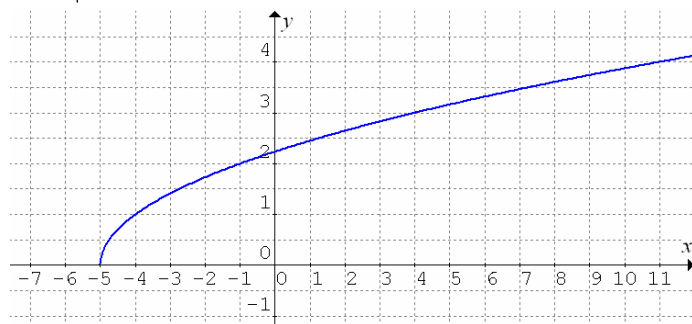


c) $k(x) = \sqrt{x+5}$,

$$k'(x) = \frac{1}{2\sqrt{x+5}} ,$$

$$k''(x) = \frac{-1}{4(x+5)^{\frac{3}{2}}}$$

D: $\{x \mid x \geq -5, x \in \mathbb{R}\}$ Zero: -5, y-int: 2.24

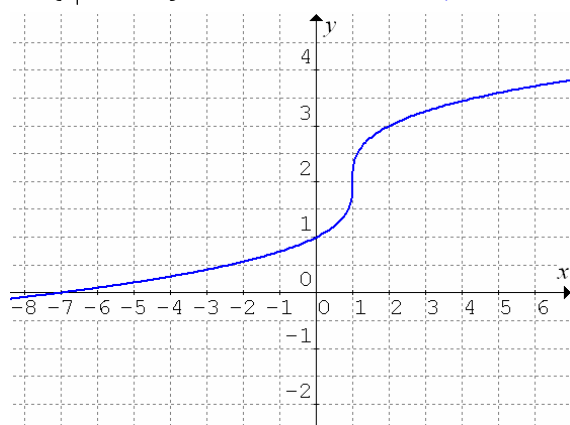


d) $j(x) = (x-1)^{\frac{1}{3}} + 2$,

$$j'(x) = \frac{1}{3}(x-1)^{-\frac{2}{3}} ,$$

$$j''(x) = \frac{-2}{9}(x-1)^{-\frac{5}{3}}$$

D: $\{x \mid x \in \mathbb{R}\}$ Zero: -7 y-int: 1 Pt. of infl. (1, 1)



e) $h(x) = x\sqrt{9-x^2}$,

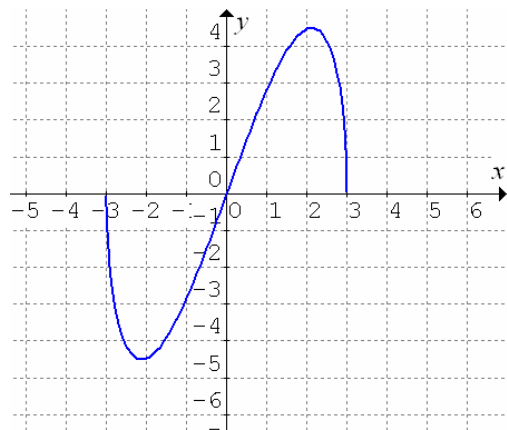
$$h'(x) = \frac{9-2x^2}{\sqrt{9-x^2}} ,$$

$$h''(x) = \frac{2x^3 - 27x}{(9-x^2)^{\frac{3}{2}}}$$

D: $\{x \mid -3 \leq x \leq 3, x \in \mathbb{R}\}$ Zero: -3, 0, 3 y-int: 0

Min (-2.12, -4.5) Max (2.12, 4.5)

Pt. of infl. (0, 0)



$$f) m(x) = \frac{(x-1)^2}{(x+1)^3},$$

$$m'(x) = \frac{-x^2 + 6x - 5}{(x+1)^4},$$

$$m'(x) = \frac{2(x^2 - 10x + 13)}{(x+1)^5}$$

$$D: \{x \mid x \neq -1, x \in \mathbb{R}\}$$

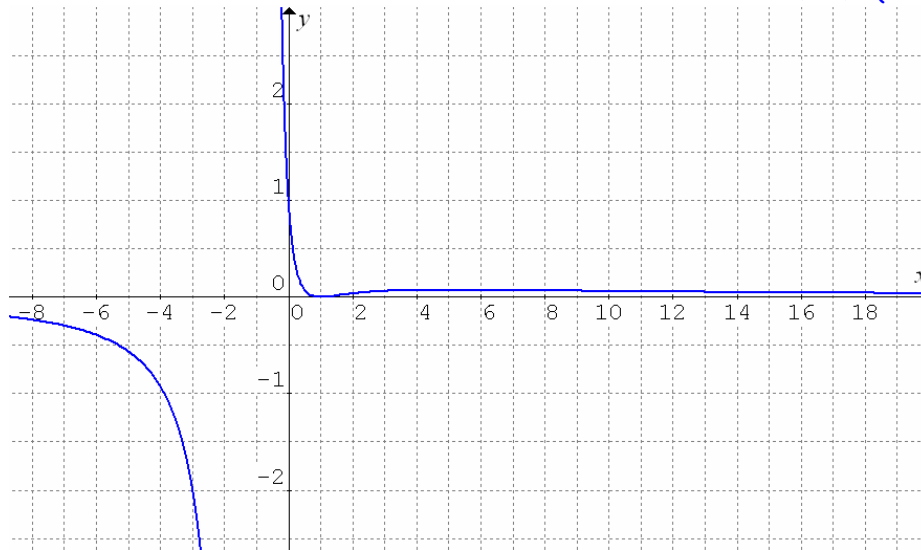
Zero: 1

y-int: 1

Min (1, 0)

Max (5, 0.07)

Pts. of infl. (1.54, 0.02) and (8.46, 0.07)



31. 79 N

32. 605 N, 513 N

33. a) N 86° E b) 1 hour 5 min

34. 93 N and 80 N

35. $\vec{a}$, $\vec{b}$, and $\vec{c}$ are unit vectors because the sum of the squares of the three components equals one.

36. a) $\frac{\sqrt{2}}{2}, \left[\frac{-1}{2}, \frac{1}{2} \right]$ b) $\frac{11\sqrt{53}}{53}, \left[\frac{44}{53}, \frac{66}{53}, \frac{-11}{53} \right]$

37. 90°

38. b) $(-5, 5, 5)$

$$\vec{r} = [0, -1, 2] + t[2, 5, -1], t \in \mathbb{R}$$

39. a) $x = 2t, y = -1 + 5t, z = 2 - t, t \in \mathbb{R}$

b) $\vec{r} = [0, 0, 1] + t[1, 0, -1], t \in \mathbb{R}$
 $x = t, y = 0, z = 1 - t, t \in \mathbb{R}$

$$\frac{x}{2} = \frac{y+1}{5} = \frac{z-2}{-1}$$

40. a) $2x - 3y - 11z + 33 = 0$ b) $y + z - 6 = 0$ c) $z - 7 = 0$

41. a) $k = \frac{1}{3}$ b) $k = 5$ or $k = -4$

42. a) $d = \frac{5\sqrt{3}}{3} \text{ units}$ b) $d = \frac{2\sqrt{14}}{7} \text{ units}$ c) $d = \frac{\sqrt{30}}{20} \text{ units}$

$$x = t$$

43. a) two planes intersect in a line: $y = -3 - 5t$, there are infinite solutions

$$z = 1 - 2t$$

b) three planes intersect at one point: $(3, -2, 4)$, there is one unique solution

c) two coincident plane with one crossing them in a line: $\vec{r} = [-18, 11, 0] + t[19, -11, 1]$,
there are infinite solutions

d) three plane are coplanar and do not intersect, there is not solution.